UNITED STATES

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

Application Type:	Provisional Patent Application
Filing Date:	March 9, 2026
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TITLE OF THE INVENTION

System and Method for Real-Time, Contextual Integration of Brand Product Placements as Narrative Solutions Within AI-Generated, Personalized Educational Content Adapted to User Belief Systems, Cultural Contexts, Developmental Profiles, and Vocabulary Acquisition Goals, With Closed-Loop Brand Engagement Analytics, Multi-Role Platform Architecture, Affiliate Referral Engine, and Guided Onboarding Framework

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FIELD OF THE INVENTION

The present invention relates generally to AI-driven educational content generation, and more particularly to: (1) dynamically integrating brand products into personalized educational narratives as organic story elements; (2) providing real-time brand engagement analytics; (3) generating content adapted across belief systems, cultural heritage, developmental profiles, and age-calibrated vocabulary; (4) managing a multi-stakeholder marketplace with five distinct roles; (5) implementing a closed-loop affiliate referral system; and (6) providing guided role-specific onboarding.

BACKGROUND OF THE INVENTION

Technical Problem

1. Static Content. Existing platforms (Duolingo, Khan Academy, ABCmouse, etc.) use pre-authored content. None personalizes narratives across religion, culture, interests, virtues, strengths, and growth areas simultaneously.

2. Disruptive Monetization. Banner ads and paywalls interrupt learning. No system integrates brand messaging *within* educational content as value-adding narrative elements.

3. No Brand Feedback Loops. Existing platforms provide only surface metrics. None gives brands: exact narrative context, comprehension questions, student pass/fail rates, or full story text.

4. Fragmented Ecosystems. No single platform serves five roles (Admin, Parent, Teacher, Student, Brand) with purpose-built portals.

5. No Cultural/Faith Personalization. No patented system generates narratives personalized for specific belief systems AND cultural contexts via LLM.

Prior Art Analysis

US 20250218307 (EZDucate): AI lessons via LLMs. No brand integration, no belief/cultural adaptation, no 60/30/10 vocabulary, no brand analytics.

CHI 2024 (POSTECH): AI storybooks via IoT. Research-only, no brands, no cultural/belief personalization, no multi-role access.

US 20250182639: AI courses. No narrative stories, no brand content, no belief/cultural personalization.

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US 11217110B2: Structured learning assets. Not narrative, no brands, no cultural personalization.

Google US 10108988B2: Video ad targeting. Standalone ads, not narrative integration.

Novelty Statement

Capability	Invention	Prior Art
AI personalized multi-chapter stories	YES	None
Brand integration as narrative elements	YES	None
Consent-gated brand content	YES	None
60/30/10 vocabulary distribution	YES	None
Closed-loop brand analytics	YES	None
Belief/cultural-aware generation	YES	None
Five-role platform	YES	None
Affiliate engine with configurable rewards	YES	None
16-level age-calibrated vocabulary	YES	None
Agentic Reach Score	YES	None
Unified multi-role authentication	YES	None
Guided onboarding with reset	YES	None

SUMMARY OF THE INVENTION

A. Content Generation Pipeline: Real-time pipeline: student profile assembly, 60/30/10 vocabulary selection, brand eligibility filtering, composite AI prompt construction, multi-chapter narrative generation.

B. Assessment Pipeline: AI-powered vocabulary evaluation, mastery tracking, Agentic Reach Score.

C. Brand Analytics: Three-layer narrative discovery, excerpt extraction, brand activation questions, engagement metrics.

D. Multi-Role Architecture: Five-tier RBAC with purpose-built portals.

E. Affiliate Engine: Auto-generated codes, configurable rewards (flat fee/percentage/credits), email confirmation, registration tracking.

F. Onboarding Framework: Role-specific wizards, localStorage persistence, tutorial reset.

G. Unified Authentication: Role selector dynamically switching between email/password and code/PIN.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Landing page

FIG. 2: Unified Login (Parent)

FIG. 3: Unified Login (Student)

FIG. 4: Admin Dashboard

FIG. 5: Word Bank Management

FIG. 6: Brand Management

FIG. 7: Affiliate Management

FIG. 8: Parent Onboarding Wizard

FIG. 9: Parent Students Tab

FIG. 10: Student Progress

FIG. 11: FAQ Section

FIG. 12: Brand Onboarding Wizard

FIGS. 13-16: Technical diagrams (to be prepared by illustrator)

FIGS. 1-12 are screenshots from a working embodiment captured March 9, 2026.

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DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

1. System Architecture

Python FastAPI async server, MongoDB (Motor), OpenAI GPT-5.2 LLM, React 18 SPA, Stripe payments, Resend email, IP geolocation, 20+ language i18n.

2. Student Profile (10 dimensions)

Name, age, grade; interest graph; 16 selectable virtues; belief system; cultural context; language (20+); parent-authored strengths; growth areas (no deficit framing); ad preferences (default-false opt-in + blocked categories); vocabulary state (banks, mastered tokens, Agentic Reach Score).

3. Vocabulary Distribution (60/30/10)

60% baseline (18 max) for reinforcement, 30% target (9 max) as primary learning, 10% stretch (3 max) for aspiration. Random shuffling ensures unique sequences per generation.

4. Age-Calibrated Vocabulary

16-level system: Pre-K (3-5 words, sight words) through Adult (unlimited, technical). All content including brand prose is developmentally appropriate.

5. Brand Eligibility Engine

Multi-factor real-time pipeline: system feature gate, guardian consent (default-false), active brand query, age filter, budget check, category block check, selection with configurable max.

6. Composite AI Prompt

Synthesizes all 10+ personalization dimensions plus brand directives into single LLM prompt. Brands integrated as organic problem-solving narrative elements.

7. Generation and Post-Processing

LLM invocation, JSON parsing, validation, persistence, impression recording, budget updates, cost logging.

8. Assessment Pipeline

Per-chapter comprehension, vocabulary assessment, AI evaluation (definition + context + spelling), mastery at 80%+, Agentic Reach Score formula.

9. Brand Analytics (Closed-Loop)

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Three-layer discovery (structured + impressions + full-text), excerpt extraction, 4-condition question identification, performance analytics, summary stats.

10. Consent Architecture

Default-false opt-in, category blocking, age filtering, budget limits, code/PIN student auth.

11. Affiliate System

AFF-XXXXXX codes, rewards (flat/percentage/credits), email confirmation, registration tracking, admin controls.

12. Brand Offers

Brand-created offers, guardian toggle/dismiss, view/click/dismissal tracking.

13. Onboarding Framework

Guardian 5 steps, Brand 5 steps, Student 4 steps. localStorage persistence. Skip/reset/re-trigger.

14. Unified Authentication

Icon-based role selector, dynamic form switching, role-specific theming, conditional registration fields.

15. Student Creation Wizard

5 steps: Basic Info, Virtues (16), Strengths/Growth, Faith/Culture, Word Banks.

16. Classroom Sessions

WebSocket-based live sessions, 6-digit codes, synchronized reading, real-time question broadcast.

17. Wallet/Payment

Stripe wallets, coupon engine, multi-currency (50+ via live exchange rates).

18. FAQ System

Role-specific content (Parent 6, Brand 6, Student 5, Teacher 5), interactive accordion.

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CLAIMS

Independent Claims

Claim 1. Method for generating personalized educational content with integrated brand placements: (a) receive request; (b) retrieve student profile (age, grade, interests, belief, culture, language, ad prefs, word banks); (c) retrieve vocabulary (baseline/target/stretch); (d) select per 60/30/10 distribution; (e) check brand authorization; (f) execute brand eligibility engine (query, filter by age/categories/budget, select); (g) construct composite LLM prompt with all dimensions + brand directives; (h) generate multi-chapter narrative; (i) store with brand data; (j) record impressions and update budgets.

Claim 2. Method for closed-loop brand analytics: (a) receive brand request; (b) identify narratives via 3-layer search (structured + impressions + full-text); (c) extract brand excerpts; (d) identify brand questions (4 conditions); (e) retrieve student interaction data; (f) compute summary stats; (g) transmit analytics.

Claim 3. System comprising: database, profile assembly, 60/30/10 vocabulary selection, brand eligibility engine, prompt construction, AI generation, impression recording, and brand analytics modules.

Claim 4. Method for affiliate referral: (a) generate AFF-XXXXXX codes; (b) automated confirmation email; (c) detect codes during registration; (d) validate; (e) record referrals; (f) calculate rewards (flat fee/percentage/wallet credits); (g) update earnings.

Claim 5. Method for guided onboarding: (a) detect first login per portal; (b) display role-specific modal guidance; (c) provide navigation; (d) persist completion state; (e) provide tutorial reset.

Dependent Claims (6-30)

Claim 6: Belief system directs LLM character behaviors/morals.

Claim 7: Cultural context directs LLM names/settings/traditions.

Claim 8: Vocabulary assessment with AI evaluation + mastery at 80%.

Claim 9: Agentic Reach Score computation formula.

Claim 10: Strengths/growth directives (no deficit framing).

Claim 11: 16-level age-calibrated vocabulary complexity.

Claim 12: Multi-layer search covers pre/post structured tracking.

Claim 13: Multi-word product name decomposition for question matching.

Claim 14: Default-false advertising opt-in.

Claim 15: Cost logging per generation event.

Claim 16: Full narrative access with brand indicators.

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Claim 17: Written response aggregation for brand review.

Claim 18: 20+ language narrative generation.

Claim 19: Multi-tier word bank ecosystem.

Claim 20: Real-time WebSocket classroom sessions.

Claim 21: Admin-configurable affiliate reward structures.

Claim 22: Per-portal customized onboarding content.

Claim 23: Unified authentication with dynamic form switching.

Claim 24: Unified registration with conditional referral field.

Claim 25: Brand promotional offers with guardian controls.

Claim 26: Contextual FAQ system with role-specific accordion.

Claim 27: Default 'every market' brand targeting.

Claim 28: Dynamic currency/geolocation (50+ currencies).

Claim 29: Time-bound referral contests with leaderboard.

Claim 30: Multi-step student creation wizard (5 steps).

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ABSTRACT

A computer-implemented system and method for generating personalized, AI-driven educational narratives that dynamically integrate brand product placements as organic, problem-solving story elements. The system assembles a multi-dimensional student profile (demographic, cultural, religious, linguistic, developmental, interest-based). A real-time brand eligibility engine applies multi-factor filtering (age, consent, categories, budget). A composite prompt synthesizes the profile, 60/30/10 vocabulary distribution, age-calibrated complexity (16 levels), and brand directives for an LLM. A closed-loop analytics pipeline provides brand partners with story excerpts, comprehension questions, and engagement metrics. Five-tier RBAC provides purpose-built portals. Additional innovations: affiliate referral engine (flat fee/percentage/wallet credits), guided onboarding wizards with reset, unified multi-role authentication, contextual FAQ, multi-step student wizard, WebSocket classroom sessions, and 50+ currency support.

OATH / DECLARATION

I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under 18 U.S.C. 1001.

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APPENDIX B: COMPETITIVE LANDSCAPE

Feature	SV	Duolingo	Khan	ABCmouse	EZDucate
AI Multi-Chapter Stories	YES	No	No	No	Partial
Belief/Cultural Adaptation	YES	No	No	No	No
Brand Narrative Integration	YES	No	No	No	No
60/30/10 Vocabulary	YES	No	No	No	No
Brand Analytics Loop	YES	No	No	No	No
Five-Role Platform	YES	No	Partial	Partial	No
20+ Language Generation	YES	Yes	No	No	No
Affiliate Referral Engine	YES	No	No	No	No
Guided Onboarding	YES	Partial	No	Partial	No
Unified Auth Interface	YES	No	No	No	No
Agentic Reach Score	YES	No	No	No	No

Document Version 2.0 | Filed March 9, 2026 | Inventor: Allen Tyrone Johnson

END OF PROVISIONAL PATENT APPLICATION

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